See Me Please Capstone Project

Model C Scoring Documentation & Handover Pack

Multi-Modal Data Fusion for Automated Accessibility & Usability Scoring

Item	Details
Prepared for	See Me Please Pty Ltd
Prepared by	CS19-1 Capstone Team
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Current status: This handover pack documents the current notebook-driven Model C/C3 scoring implementation, including Proposal 1, Proposal 2, manual review validation, Tester Quality, rollup outputs, and supporting data dictionary artefacts prepared for client handover and future development.

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1. Executive Summary

This handover pack documents the current notebook-driven Model C/C3 scoring implementation prepared for See Me Please. The current deliverable includes Model C/C3 refinement, Proposal 1 graduated deductions, Proposal 2 time moderation, manual review validation outputs, rollup outputs, presentation assets, a separate Tester Quality Rating prototype, and a supporting data dictionary package.

> Core position: the deliverable is a notebook-driven Model C/C3 scoring handover package that implements Proposal 1 graduated deductions, Proposal 2 time moderation, manual review validation, and a separate Tester Quality Rating prototype. It is a working analytical prototype and handover package rather than a production-integrated SMP scoring system.

The document is intended to support client review, future maintenance, and future development decisions. It is not a production deployment manual. Instead, it explains the implemented scoring logic, the repository workflow, the output artefacts, the audit trail, the current limitations, and the recommended next steps before any broader operational use.

1.1 What the team has delivered

- A clarified scoring problem statement focused on reducing over-reliance on raw self-reported survey scores while respecting participant voice.
- A comparison-informed decision to focus on Model C/C3 as the most practical, transparent, and feasible current MVP direction.
- A working notebook-driven Model C/C3 implementation using survey-task alignment, insight deduplication, L1/L2/L3 scoring, confidence thresholding, cohort-aware weighting, and explanation fields.
- Proposal 1 implemented as graduated medium-friction deductions with explicit constants, floors, and score explanations.
- Proposal 2 implemented as an L3 usability time-moderation layer using participant-normalised time evidence, matched/unmatched diagnostics, and row-level audit outputs.
- A separate Tester Quality Rating prototype implemented as an independent third score for feedback-quality and evidence-quality review.
- A manual review validation package comparing raw survey scores, Model C before time moderation, and Proposal 2 after time moderation against curated transcript-aligned review evidence.
- A practical handover structure that documents implemented components, available evidence files, known limitations, and future development priorities for SMP.

1.2 Current Implementation Status

Component	Current status	Evidence / output artefact	Client-facing interpretation
Model C / C3 scoring	Implemented	Notebook 01; outputs/proposal1_rows.csv; outputs/model_c3_proposal1_results.xlsx	Working notebook-based analytical scoring implementation prepared for handover and future development.
Proposal 1 graduated deduction	Implemented	Notebook 01; proposal1 output files; explanation fields	Medium-friction scoring is less cliff-edge than under a hard-cap-only approach, while severe rules remain bounded.

Component	Current status	Evidence / output artefact	Client-facing interpretation
Proposal 2 time moderation	Implemented for L3 usability where mapped time evidence exists	Notebook 02; proposal2_integrated_l3_time_score.csv; proposal2_time_audit_log.csv	Time moderation is active within current scope and is auditable, but remains suitable for future calibration.
Manual review validation	Implemented as an 18-case validation package	Notebook 03; manual_review_outputs folder	Current validation evidence supports calibration and interpretation, not definitive ground-truth claims.
Tester Quality Rating	Implemented as a standalone third-score prototype	Notebook 04; tester_quality_outputs folder	Independent evidence-quality signal that does not alter Accessibility or Usability scores.
Retry behaviour	Explored as a future operational signal, not a direct scoring rule	Current scoring notebooks; validation notes	Useful as supporting evidence only until a formal extraction and weighting method is agreed.
Rollup outputs	Implemented	Notebook 06; proposal2_rollup_task.csv; proposal2_rollup_participant.csv; proposal2_rollup_project.csv	Provides task-, participant-, and project-level summaries for review and reporting.
Presentation assets	Implemented	Notebook 05; outputs/presentation_assets/	Useful for reporting and visual explanation, but not a substitute for row-level audit files.
Data dictionary package	Included	final_data_dictionary_package/	Provides the field-level reference for future maintenance and interpretation.
Production integration	Future work	No production integration artefact in current package	The deliverable supports future development rather than direct replacement of SMP's production system.

The current repository already includes the main notebooks, outputs folder, validation artefacts, Tester Quality artefacts, rollup files, and a separate data dictionary package. Items that depend on future SMP operating decisions are summarised in the limitations, recommendations, and Appendix D artefact register.

2. Document Purpose, Scope and Evidence Status

2.1 Purpose of this handover pack

This documentation is intended as a practical handover resource for See Me Please. It is written for three audiences:

1. Product and client stakeholders who need to understand what the scoring model is doing and why.
2. Technical teams or future capstone teams who may continue the scoring implementation.
3. Reviewers who need to audit whether score adjustments are explainable, conservative, and aligned with the participant experience.

2.2 Scope of this document

In scope	Out of scope / deferred
Notebook-driven Model C/C3 scoring handover package	Direct production deployment inside the SMP platform
Accessibility and Usability score logic	Full production integration, orchestration, and monitoring
Proposal 1 graduated deduction logic	Large-scale statistical calibration and confidence intervals
Proposal 2 time moderation where mapped evidence exists	Complete end-to-end time coverage for every participant and task
Manual review validation package	Definitive ground-truth claims
Tester Quality Rating prototype	Production psychometric or reliability model
Data dictionary, audit artefacts, and repository run guidance	Final client-approved reporting templates unless separately prepared

Appendix D lists the accompanying artefacts provided with the handover package and notes any items that depend on future packaging or reporting decisions.

2.3 Evidence status

Evidence category	Status in current package	Documentation treatment
Project brief and deliverable expectations	Available	Used to frame purpose, scope, and client-facing interpretation.
SMP success criteria	Available	Used to frame explainability, auditability, and scoring-governance requirements.
Meeting minutes and progress records	Available	Used as contextual background only; current implementation details are drawn from the repository package.

Evidence category	Status in current package	Documentation treatment
Client presentation decks	Available	Used to support background explanation and presentation-asset interpretation.
Implementation notebooks	Available	Treated as the primary technical source for current scoring behaviour.
Current CSV/XLSX outputs	Available in current repository/package	Used to support output, auditability, validation, and reporting sections.
Validation package files	Available in current repository/package	Used as current calibration evidence, with anonymisation recommended for embedded examples.
Repository README / run guidance	Available in current repository/package	Used to explain practical notebook order and future reproducibility.

3. Client Problem and Success Criteria

3.1 Client problem

The central problem is that See Me Please collects rich evidence from accessibility and usability testing, but the current scoring process relies heavily on participant self-reported survey scores. Survey scores are valuable because they represent a participant's own reflective judgement, but they can also be influenced by positive bias, survey fatigue, normalisation of barriers, question interpretation, and individual response style.

The scoring model therefore needs to hold multiple evidence streams together: structured survey responses, observed task outcomes, LLM-derived transcript insights, and behaviour evidence such as time to complete or repeated attempts. The objective is not to replace the participant's voice with an algorithm. The objective is to produce scores that are more credible, explainable, and auditable than raw survey averages alone.

3.2 Deliverable expectations from the project brief

Brief expectation	How this documentation addresses it
Documented scoring algorithms for Accessibility, Usability, and Tester Quality	Documents the current Accessibility and Usability scoring logic and the implemented independent Tester Quality prototype, while keeping Tester Quality separate from ACC/USA.
Functional prototype demonstrating algorithm performance	Documents the notebook-driven Model C/C3 implementation, Proposal 1, Proposal 2, rollups, and current outputs.
Validation framework	Documents the implemented manual-review validation package and its cautious interpretation.
Technical documentation	Provides repository workflow, input assumptions, scoring logic, output structure, auditability, and maintenance references.

Brief expectation	How this documentation addresses it
Research and recommendations report	Provides practical limitations, risk analysis, and future-development priorities.
Clear documentation enabling future maintenance	Provides a handover pack that can guide review, reproduction, and future productionisation decisions.

3.3 SMP success criteria translated into documentation requirements

Success criterion	Practical documentation requirement
1. More credible than the current MVP	Show how scores relate to observed friction and do not rely solely on survey averages.
2. Distinguish accessibility and usability	Document separate ACC/USA streams, friction routing, and cohort-aware weights.
3. Reflect severity of barriers	Document blocker, severe, high, medium and low friction handling.
4. Handle survey data honestly	Show survey as a ceiling/anchor, not as ground truth and not as something routinely discarded.
5. Do not over-rely on LLM signal	Document confidence threshold, LLM limitations, and conservative treatment of LLM-derived insights.
6. Conservative in correcting participant voice	Show that downward corrections require strong evidence and remain auditable.
7. Explainable to participants	Include explanation columns and narrative reasons for score changes.
8. Implementable within current capacity	Avoid over-complexity; prioritise rule-based, auditable MVP logic.
9. Foundations for validation and refinement	Keep intermediate outputs, audit logs, and flags for later calibration.

4. Current Repository and Notebook Workflow

4.1 Repository workflow and notebook roles

The current repository is organised around a notebook-driven workflow rather than a single application entry point. Each notebook has a defined role in the scoring, moderation, validation, reporting, or packaging process, and together they form the handover package for future SMP review and development.

This structure is appropriate for the current deliverable because it keeps each major analytical step transparent, reviewable, and separable. It also makes it easier to inspect intermediate outputs before any future productionisation work is planned.

4.2 Practical execution sequence

Notebook / file	Purpose	Key inputs	Key outputs	Role in handover package
01_run_model_c_proposal1.ipynb	Base Model C / Proposal 1 scoring	Raw survey export; insights workbook	proposal1_rows.csv; proposal1_df_insights.csv; model_c3_proposal1_results.xlsx; Proposal 1 summaries	Primary scoring implementation for ACC/USA before Proposal 2.
02_proposal2_time_integration.ipynb	Proposal 2 time moderation for L3 usability	Proposal 1 outputs; raw Proposal 2 time signal file; reference l3_score_before_time.csv	proposal2_integrated_l3_time_score.csv; proposal2_time_audit_log.csv; integration summary; unmatched diagnostics	Implements current time moderation scope and audit trail.
03_manual_review_validation.ipynb	Manual-review validation package	Proposal 2 integrated output; task segmentation; transcript segments	manual_review_cases.csv; validation summary; error details; findings and case examples	Provides current validation evidence for review and calibration.
04_tester_quality_rating.ipynb	Tester Quality Rating prototype	Raw survey export; Proposal 2 outputs; insights; manual-review outputs; transcript segments	tester_quality_scores.csv; tester_quality_features.csv; tester_quality_summary.csv; methodology and examples	Implements the independent third-score prototype.
05_presentation_assets.ipynb	Presentation and reporting assets	Proposal 2 outputs; validation summary; Tester Quality outputs	PNG charts and pipeline visuals	Supports client review and presentation packaging.
06_proposal2_rollup_outputs.ipynb	Proposal 2 rollups	proposal2_integrated_l3_time_score.csv	proposal2_rollup_task.csv; proposal2_rollup_participant.csv; proposal2_rollup_project.csv	Provides summary-level reporting artefacts.

The practical run order is Notebook 01, then 02, then 03, then 04, then 05, and finally 06. This order reflects the dependency chain from base scoring through time moderation, validation, Tester Quality, presentation assets, and rollup outputs.

> Workflow note: the repository is organised as a staged notebook package. Notebook 01 establishes the base scoring outputs, Notebook 02 adds time moderation, Notebook 03 provides manual-review validation, Notebook 04 produces Tester Quality outputs, Notebook 05 prepares presentation assets, and Notebook 06 produces rollups.

Repository execution order and run guidance are included in README_How_To_Run.md. The handover package therefore includes both the notebook workflow and the supporting run instructions for future review.

5. Data Inputs, Assumptions and MVP Boundaries

5.1 Input streams used or considered

Input stream	Current role in handover package	Notes / assumptions
Survey responses	Starting score anchor, participant metadata, expectation responses, and open-ended feedback inputs.	Survey responses remain important evidence but are not treated as self-sufficient final truth.
LLM-derived friction / insight evidence	Supports friction type, severity, confidence, and task-level evidence routing.	Treated as supporting evidence with a confidence threshold, not as ground truth.
Task mapping	Links survey questions, Model C tasks, and Proposal 2 task references.	Mapping quality matters for downstream scoring and diagnostics.
Cohort mapping	Supports current comprehension weighting logic.	Used as implementation metadata and suitable for future review.
Time evidence for Proposal 2	Supports L3 usability moderation where mapped evidence exists.	Coverage is partial and unmatched diagnostics should be preserved.
Manual review data	Provides current calibration and validation evidence.	Should be interpreted cautiously and anonymised in client-facing examples.
Tester Quality feature inputs	Supports feedback-depth, coverage, consistency, and engagement calculations.	Used only for the separate Tester Quality prototype.
Rollup inputs	Support task-, participant-, and project-level summaries.	Derived from current row-level outputs rather than replacing them.
Anonymisation and privacy assumptions	Controls how examples should be embedded in client-facing material.	Any client-facing sample outputs should use anonymised or client-approved data.
Matching assumptions	Support joins across participant, task, survey, time, and validation evidence.	Where matching is incomplete, diagnostics should be kept visible.

The current implementation uses multiple input groups: survey scores and participant metadata, LLM-derived friction/insight evidence, task and cohort mapping inputs, Proposal 2 time evidence, transcript segment files used for validation, Tester Quality feature inputs, and downstream rollup inputs. The final_data_dictionary_package should be treated as the detailed field-level reference for future maintenance.

A concise sample of the included data dictionary is shown below. The full final_data_dictionary_package accompanies the handover package and provides the detailed field-level reference.

Field	Current meaning in the package	Used in
participant_key	Canonical participant join key used in processed scoring; raw sources use multiple identi…	All scoring, Proposal 2, Manual Review, Tester Quality
primary_cohort	Primary or raw cohort/disability group information used for weighting, segmentation, or r…	Model C / Proposal 1 / Tester Quality
raw_survey_score	Participant self-reported survey score or raw answer payload used as the ceiling/baseline.	Model C / Proposal 1 / Manual Review / Proposal 2
survey_question	Survey prompt or canonical question metadata.	Model C / Proposal 1 / Raw Survey / Output Documentation

5.2 MVP boundaries

The handover package assumes that participant, task, and survey-question matching are handled carefully and that any client-facing examples use anonymised or client-approved data. It also assumes that some Proposal 2 participant/task matches will remain incomplete, and that unmatched diagnostics should be preserved rather than hidden.

- The scoring package prioritises the Accessibility and Usability scores before the separate Tester Quality prototype.
- The current implementation uses survey evidence, friction evidence, task outcomes, time evidence, and transcript-aligned validation artefacts before any richer clickstream or full video automation.
- Any client-facing sample outputs should use anonymised or client-approved data, especially where participant email, transcript content, or task-aligned excerpts appear in repository outputs.
- The delivered output structure includes intermediate scoring steps and explanation fields so future teams can validate and refine the implementation.

6. Scoring Scenario Context: A, B, C and D

Scenario	Core logic	Role in this project	Main risk / limitation
Scenario A: Raw survey baseline	Directly maps participant survey ratings into scores.	Baseline for showing why raw survey averages are insufficient.	Can be overly positive and can hide severe friction.
Scenario B: Current/manual expert process	Manual overwriting or interpretation using expert review and LLM-analyzed insights.	Reference for why repeatable algorithms are needed.	Not formalised, not scalable, and hard to reproduce consistently.

Scenario	Core logic	Role in this project	Main risk / limitation
Scenario C/C3: Survey-ceiling / guardian model	Starts from survey score and applies downward evidence-based caps or deductions.	Primary current MVP direction and focus of this handover pack.	Can be too harsh if hard caps are triggered by moderate friction or incomplete evidence.
Scenario D: Evidence-first reconciliation model	Generates observed scores from evidence and reconciles with survey scores.	Comparison reference and long-term design inspiration.	More complex, heavier governance burden, and harder to validate in current timeline.

6.1 Why Model C/C3 was prioritised

- It is easier to explain to customers and participants because the participant survey score remains visible as the starting point.
- It is conservative: the model generally reduces scores only when evidence suggests the survey score is too positive.
- It is feasible within the current data and timeline because it does not require a fully independent observed scoring engine.
- It aligns with client feedback that the model should be less wrong, practical, auditable, and not overly complex.
- Its limitations are concrete and refinable, especially around hard caps, time moderation, and additional behavioural signals.

7. Model C/C3 Overview

7.1 Design principles

Principle	Meaning in Model C/C3
Survey as starting point	The participant's score remains the starting ceiling and is not discarded wholesale.
Scores only go down	Evidence can reduce or cap a score, but the model avoids inflating scores above the participant's stated rating.
Severe barriers must matter	Blocked tasks, severe accessibility friction, or high accessibility failures cannot be masked by positive survey responses.
LLM signal is limited	LLM insight supports scoring but should not act as the sole or dominant determinant without confidence and corroboration.
Intermediate outputs matter	The model should keep L1/L2/L3 scores, final scores, explanations, and audit fields for review.

The current implementation remains a survey-anchored, evidence-based adjustment model. It keeps the participant's survey score as the starting reference, separates accessibility and usability, applies documented adjustment rules through L1/L2/L3 layers, and records plain-language explanation fields so that score changes remain transparent.

7.2 Conceptual pipeline

Step	Pipeline stage	Purpose
1	Load and standardise inputs	Load survey data, participant metadata, task mappings, cohort labels, and insight evidence.
2	Normalise survey scores	Convert 1–5 survey responses to 0–100 reference scores.
3	Map survey questions to Model C tasks	Create task-level rows that can be scored consistently.
4	Route and deduplicate friction evidence	Assign evidence to AT, COMP, and USA streams and reduce duplicate rows.
5	Apply L1 rules	Handle blocker/severe accessibility behaviour and high AT caps.
6	Apply L2 accessibility logic	Generate Accessibility score using AT and weighted comprehension evidence.
7	Apply L3 usability logic	Generate Usability score using USA, weighted comprehension evidence, and expectation moderation.
8	Apply Proposal 1 graduated deductions	Reduce cliff-edge medium-friction behaviour while preserving stream-specific floors.
9	Apply Proposal 2 time moderation	Moderate L3 usability downward where mapped participant-normalised time evidence supports it.
10	Generate outputs and audit artefacts	Produce row-level outputs, explanations, audit logs, validation outputs, Tester Quality outputs, and rollups.

At a high level, the workflow standardises survey inputs, maps survey questions to Model C tasks, routes friction evidence into the appropriate streams, applies L1/L2/L3 scoring, then applies Proposal 2 time moderation to L3 usability only where mapped time evidence exists. Validation, rollups, and Tester Quality are generated as separate downstream layers.

Visual workflow assets are included in the presentation assets package, including implementation_pipeline_diagram.png and model_c_integration_pipeline.png. These can be used in presentation material alongside this handover document where required.

8. Current Model C/C3 Scoring Logic

8.1 Friction stream routing

Original friction type	Scoring stream	Interpretation
accessibility	AT	Direct accessibility / assistive technology barrier.
comprehension	COMP	Understanding, language, content interpretation or instruction-related barrier.
content_not_found	USA	Usability issue where participant could not locate required content.
confidence	USA	Usability-related uncertainty or lack of confidence.
excessive_effort	USA	Usability issue due to disproportionate effort.
unexpected_behaviour	USA	Interface behaviour did not match participant expectation.
unresponsive_interface	USA	Interface did not respond as expected.

The current implementation routes accessibility barriers into the AT stream, comprehension and language-related barriers into the COMP stream, and usability-related barriers into the USA stream. This separation supports different treatment of accessibility and usability while still allowing comprehension friction to influence both where appropriate.

8.2 Confidence threshold

The implementation uses a confidence threshold of 0.70. Insights below this threshold do not trigger the main implemented caps or graduated deductions. This reflects the success criterion that LLM-derived signal is treated conservatively as supporting evidence rather than ground truth.

```
CONFIDENCE_THRESHOLD = 0.70
```

8.3 Cohort-aware comprehension weights

Cohort code	ACC weight	USA weight	Implementation interpretation
ESL	1.0	0.4	Comprehension friction is treated primarily as accessibility-related in the current implementation.
DF	1.0	0.4	Comprehension friction is treated primarily as accessibility-related in the current implementation.
ND	1.0	0.4	Comprehension friction is treated primarily as accessibility-related in the current implementation.

Cohort code	ACC weight	USA weight	Implementation interpretation
BL	0.5	0.9	Comprehension friction contributes to both accessibility and usability, weighted more heavily toward usability.
LV	0.5	0.9	Comprehension friction contributes to both accessibility and usability, weighted more heavily toward usability.
OL	0.6	0.8	Mixed weighting in the current implementation.
AB	0.4	1.0	Comprehension friction is treated primarily as usability-related in the current implementation.
OT	0.4	1.0	Default or other-group weighting in the current implementation.

The current implementation uses cohort-aware comprehension weighting as internal scoring metadata. These weights are active in the code and are documented in this handover pack. Whether cohort labels themselves should be surfaced in future client-facing reporting remains a presentation and governance decision for SMP.

The cohort codes currently used in the implementation are ESL, DF, ND, BL, LV, OL, AB, and OT. These codes are sufficient for documenting the current logic and are described here as implementation metadata rather than as final client-facing reporting labels.

8.4 L1 categorical rules

Rule	Current logic	Rationale
L1.1 ceiling	Start from the participant's survey-derived ceiling score.	Respect participant voice and retain survey response as the scoring anchor.
L1.2 blocker / severe AT	If blocker or severe accessibility friction exists, score becomes 0.	Categorical exclusion or severe access failure must not be masked by a high survey score.
L1.3 high AT	One high AT event caps score at 20; two or more high AT events cap score at 10.	High accessibility barriers require strong downward correction.
L1.4 confidence threshold	Only insights with confidence >= 0.70 can trigger rules.	Avoid over-reliance on weak LLM signal.
L1.5 confident-positive floor	If survey score >= 75, expectations met/exceeded, and no high/severe friction, restore score to at least 55.	Prevents medium-friction logic from over-penalising otherwise positive and non-severe experiences.

L1 applies the initial categorical logic. Survey scores are normalised from 1–5 into a 0–100 reference score, severe or blocker accessibility evidence can reduce the score to 0, and high AT evidence can apply stronger caps before the downstream layers are calculated.

> Proposal 1 deliberately leaves L1 categorical rules unchanged. Blocker and severe access failures are treated as safety rules rather than proportional penalties.

L2 generates the accessibility score using accessibility-stream evidence and weighted comprehension evidence. L3 generates the usability score using usability-stream evidence, weighted comprehension evidence, and the expectation-response moderator. The final outputs also retain explanation fields and a conservative positive-floor safeguard where the participant's overall evidence remains positive and non-severe.

8.5 L2 accessibility logic

The confidence threshold for triggered deductions is 0.70. Below this threshold, insight evidence may remain contextually informative but does not automatically trigger the implemented graduated-deduction or cap logic.

Evidence condition	Effect under current/enhanced logic
Medium accessibility friction	Graduated deduction using base penalty 20 and floor 30.
Low accessibility friction	Caps may apply at 70 for one or more events and 60 for three or more events.
Severe/high comprehension friction	Weighted cap pressure toward 40 depending on ACC cohort weight.
Medium comprehension friction x1-x2	Graduated deduction using base penalty 15 and floor 50, then cohort weighting.
Medium comprehension friction x3+	Hard cap 40 retained and stacking logged for future review.
Low comprehension friction	Weighted cap pressure toward 70 or 60 depending on count.
Expectations exceeded	May restore accessibility score up to a protected floor based on accessibility friction floor.

8.6 L3 usability logic

The expectation moderator remains part of the implemented logic. In the usability stream, a response of "not met" reduces the score by multiplication, while "fell well below" reduces it more strongly. In addition, a positive-floor safeguard protects certain non-severe, high-survey-score cases from being over-penalised.

Evidence condition	Effect under current/enhanced logic
Severe usability or 3+ high usability friction	Hard cap at 50.
High + medium usability friction	Hard cap at 60.
3+ medium usability friction	Graduated deduction using base penalty 25 and floor 70.
Medium + low usability friction	Graduated deduction using base penalty 12 and floor 75.
3+ low usability friction	Cap at 80.
Comprehension friction	Weighted using USA cohort weights, with medium friction graduated for x1-x2 and hard cap retained for x3+.
Expectation response: not met	Score multiplied by 0.80.
Expectation response: fell well below	Score multiplied by 0.60.

The output files preserve L1, L2, L3-before-time, final accessibility/usability scores, and explanation fields where available. This is a key part of the auditability design and remains important for any future productionisation work.

9. Model C/C3 Limitations Identified

The main design boundaries arise from the same qualities that make the model transparent: survey scores remain an important reference point, time evidence is partial and sometimes noisy, cohort weighting is heuristic rather than clinically calibrated, and LLM-derived signals are treated as supporting evidence rather than definitive truth. These boundaries do not invalidate the implementation, but they matter for interpretation and future development planning.

Limitation	Why it matters	Current response
Hard caps can be too harsh	Medium friction should not always produce cliff-edge score drops.	Proposal 1 replaces selected hard caps with graduated deductions.
Survey vs observed evidence mismatch	Participants may rate a journey positively despite hidden friction or normalised barriers.	Model C uses evidence-based downward correction and Proposal 2 adds behavioural moderation.
LLM insight incompleteness	Some barriers are not verbalised or are difficult to interpret from transcript evidence alone.	LLM-derived evidence is thresholded and treated as supporting evidence.
Extreme outputs require explanation	Very low scores need transparent justification.	Explanation fields and audit artefacts are included in the current outputs.
Time evidence can be noisy	Longer duration may reflect personal pace, assistive technology use, exploration, or genuine friction.	Proposal 2 uses participant normalisation, clipping, and diagnostics, but remains suitable for calibration.
Retry remains ambiguous	Repeated behaviour may reflect exploration, screen-reader navigation, or genuine difficulty.	Retry remains descriptive/supporting rather than a direct scoring rule.
Data inconsistency affects matching	Task names, participant identifiers, and timing references need careful alignment.	Data dictionary references and unmatched diagnostics are included in the package.

10. Enhancement Proposal 1: Graduated Deductions

10.1 Problem addressed

Proposal 1 addresses the “cliff-edge” behaviour of hard caps for manageable or medium friction. In the original hard-cap approach, two participants experiencing the same medium friction could receive very different score drops simply because they started with different survey scores. For example, a survey score of 90 capped at 30 loses 60 points, while a survey score of 52 capped at 30 loses 22 points. This is difficult to justify to stakeholders when the observed friction is identical.

10.2 New scoring mechanic

Proposal 1 is implemented as a stream-specific graduated deduction layer for selected medium-friction cases. Instead of forcing a participant directly down to a hard cap whenever medium friction appears, the implementation

applies a confidence-weighted deduction and then protects a stream-specific floor so that the score reduction remains bounded and explainable.

```
confidence_weight = (confidence - 0.70) / 0.30
deduction = base_penalty * confidence_weight
final_score = max(floor, current_score - deduction)
```

> Low confidence produces little or no deduction. High confidence produces a larger deduction, but the floor keeps the behaviour auditable and bounded.

This is the main mechanism used to reduce the earlier cliff-edge behaviour in medium-friction scenarios while preserving stricter treatment for blocker, severe, and high-friction cases.

10.3 Proposal 1 rule changes

Stream	Trigger	Original C3 behaviour	Proposal 1 behaviour	Status
AT	Medium accessibility friction	Hard cap 30	Base penalty 20; floor 30	Changed
COMP	Medium comprehension x1-x2	Weighted cap 50	Base penalty 15; floor 50; then weighting	Changed
COMP	Medium comprehension x3+	Weighted cap 40	Hard cap 40 retained; stacking logged	Monitoring
USA	Medium usability x3+	Hard cap 70	Base penalty 25; floor 70	Changed
USA	Medium + low usability	Hard cap 75	Base penalty 12; floor 75	Changed
Blocker / severe / high accessibility	Critical barriers	Hard cap or 0	Unchanged	Safety rule retained

In the current code, the implemented graduated deduction settings are: medium AT = base penalty 20, floor 30; medium COMP = base penalty 15, floor 50; medium USA mixed/combined friction = base penalty 12, floor 75; and stacked medium USA friction = base penalty 25, floor 70. These deductions scale only when confidence is at least 0.70.

10.4 Implementation status

- The Proposal 1 implementation is evidenced by the base scoring notebook and its output files. These include row-level scoring outputs, the processed insight table used in scoring, workbook exports, and task/cohort/project summaries that can be reviewed independently.
- The key Proposal 1 artefacts are outputs/proposal1_rows.csv, outputs/proposal1_df_insights.csv, outputs/model_c3_proposal1_results.xlsx, outputs/proposal1_task_summary.csv, outputs/proposal1_cohort_summary.csv, and outputs/proposal1_project_summary.txt.
- These outputs show the survey-derived reference score, the triggered scoring logic, the intermediate L1/L2/L3 values, the final ACC/USA scores, and the explanation text used to support later review.
- Proposal 1 therefore changes score behaviour in a targeted way: medium friction becomes less punitive than under a pure hard-cap approach, while severe and blocker behaviour remains clearly bounded by stronger safety rules.
- Comprehension stacking events logged: 0 in the provided notebook output.

The current repository includes the Proposal 1 output workbook and related CSV/TXT summaries as part of the handover package.

10.5 Example explanation output

Output field	Client-facing purpose
acc_explanation	Provides a plain-language explanation of why the Accessibility score stayed the same or changed.
usa_explanation	Provides a plain-language explanation of why the Usability score stayed the same or changed.

The explanation fields are designed to support review by both technical and non-technical stakeholders. Where a score changed, the explanation can reference the triggered friction condition, the relevant confidence level, the applicable floor, and whether a cohort weight influenced the result.

Proposal 1 sample rows below are taken from the delivered repository outputs so the document reflects implemented behaviour rather than illustrative wording.

A representative anonymised sample is shown below. The full Proposal 1 output package is included in the accompanying handover artefacts, including outputs/proposal1_rows.csv, outputs/proposal1_df_insights.csv, outputs/model_c3_proposal1_results.xlsx, outputs/proposal1_task_summary.csv, outputs/proposal1_cohort_summary.csv, and outputs/proposal1_project_summary.txt.

Participant	Task	Survey reference	ACC final	USA final	ACC explanation (summary)	USA explanation (summary)
Participant A	AAMI Website	80.0	80.0	80.0	No friction rules changed this participant's ACC score. Final score: 80 (from survey score 80).	No friction rules changed this participant's USA score. Final score: 80 (from survey score 80).
Participant B	AAMI Website	80.0	74.1	67.0	This participant's ACC score of 74 reflects a reduction from their survey score of 80 due to: 1 medium compre…	This participant's USA score of 67 reflects a reduction from their survey score of 80 due to: 1 medium + 1 lo…
Participant C	AAMI Website	60.0	51.0	56.4	This participant's ACC score of 51 reflects a reduction from their survey score of 60 due to: 2 medium compre…	This participant's USA score of 56 reflects a reduction from their survey score of 60 due to: 2 medium compre…

11. Enhancement Proposal 2: Time Moderation Integration

11.1 Problem addressed

Proposal 2 addresses the weakness of raw task-duration comparison. A participant may take longer because they are generally slower, because they use assistive technology, or because a specific task contains hidden friction. Comparing raw duration directly to a task median can therefore create false-positive friction. Proposal 2 reduces this risk by normalising task duration against each participant's own speed profile before applying time moderation.

11.2 Time moderation pipeline

Step	Process	Purpose
1	Load Proposal 1 outputs and raw time signal data	Bring together L3-before-time scores and task timing evidence.
2	Task and participant alignment	Match time rows to Model C rows through canonical keys and participant mapping.
3	Recommended duration reference	Use task-level recommended durations as part of participant speed normalisation.
4	Participant speed-factor calculation	Estimate general participant pace before task-level comparison.
5	Speed-factor clipping (0.7–1.5)	Prevent extreme pace outliers from destabilising the moderation step.
6	Normalised duration calculation	Convert raw time into participant-normalised behavioural evidence.
7	Task-median comparison	Compare each normalised task duration to the task-level normalised median.
8	Time-cap decision and audit output	Apply L3 usability cap where supported and record the explanation/audit fields.

Proposal 2 is implemented as a time-based moderation layer for L3 usability. It uses participant-normalised time evidence rather than raw duration alone, and it does not alter accessibility scores. It applies only where mapped time evidence exists for the relevant participant-task combination.

The current implementation uses recommended task durations, participant speed-factor clipping, and normalised task-median comparison to decide whether a time-based cap should apply. The time signal therefore acts as a behavioural moderation layer rather than a raw-duration penalty.

```
speed_factor = actual_duration / recommended_duration
speed_factor = clip(speed_factor, 0.7, 1.5)
normalised_duration = raw_duration / speed_factor
l3_score_after_proposal2_time = min(l3_score_before_time, proposal2_time_cap)
```

- Proposal 2 can reduce the usability score or leave it unchanged, but it must not increase a score. This is enforced in the integration quality checks.
- L1 and L2 remain unchanged, and accessibility outputs remain those produced by Proposal 1 / Model C logic.
- The Proposal 2 layer therefore affects current scope only at L3 usability where time evidence can be matched confidently.
- The current time signal bands are: ratio below 1.2 = normal, 1.2–1.5 = mild time friction, 1.5–2.0 = medium time friction, and 2.0 or above = strong time friction. Where caps are triggered, the implemented thresholds are 50 at 2x–3x, 40 at 3x–4x, and 25 at 4x+.
- The participant speed factor is clipped between 0.7 and 1.5 to reduce the effect of extreme pace outliers. Matched and unmatched participant/task diagnostics are also retained so that missing time links remain visible.

11.4 Existing time cap bands

Normalised time ratio	Time cap	Interpretation
2x-3x task benchmark/median	Cap 50	Moderate-to-high time friction.
3x-4x task benchmark/median	Cap 40	High time friction.
4x+ task benchmark/median	Cap 25	Severe time friction.

Time is a useful but still noisy behavioural signal. It can reflect genuine friction, slower reading, assistive technology use, environmental context, or a participant's style of speaking and exploration. For that reason, the current thresholds should be treated as implemented settings that remain suitable for future calibration.

11.5 Current Proposal 2 Integration Results

Metric	Current result
Model C rows processed	136
Rows matched to time evidence	95
Rows with time cap triggered	14
Rows with L3 usability score changed after time moderation	12
Score-increase violations	0
Unmatched diagnostics	Preserved in dedicated participant/task files

The current Proposal 2 integration outputs show 136 Model C rows processed, 95 rows matched to Proposal 2 time evidence, 14 rows where a time cap was triggered, and 12 rows where the L3 usability score changed. The current artefact set includes outputs/l3_score_before_time.csv, outputs/proposal2_integrated_l3_time_score.csv, outputs/proposal2_time_audit_log.csv, outputs/proposal2_integration_summary.csv, outputs/proposal2_time_signal_result_recomputed_for_integration.csv, outputs/proposal2_unmatched_participants.csv, outputs/proposal2_unmatched_tasks.csv, outputs/proposal2_human_readable_summary.txt, outputs/proposal2_rollup_task.csv, outputs/proposal2_rollup_participant.csv, and outputs/proposal2_rollup_project.csv.

A representative anonymised sample is shown below. The full Proposal 2 moderation package is included in the accompanying handover artefacts, including the integrated row-level output, time-audit log, integration summary, recomputed time-signal file, unmatched diagnostics, and rollup outputs.

Participant	Task	L3 before time	Normalised ratio	Time signal	Cap	L3 after time	Audit explanation (summary)
Participant A	AAMI Website	90.1	2.51	Strong time friction	50	50.0	2.51x median; cap reduced L3 to 50.
Participant B	AAMI Website	100.0	3.56	Strong time friction	40	40.0	3.56x median; stronger cap reduced L3 to 40.

12. Tester Quality Rating

12.1 Purpose and position in the handover package

Tester Quality Rating is implemented as a separate notebook-based prototype and independent third score. It does not modify the Accessibility or Usability scores. Instead, it provides an additional view of the quality and reliability of tester feedback and supporting evidence.

> Interpretation note: Tester Quality Rating is an independent third score for assessing the reliability, specificity, coverage, and usefulness of tester feedback. It does not modify Accessibility or Usability scores.

The current implementation is best understood as a heuristic scoring prototype rather than a production psychometric model. It is intended for review prioritisation, evidence-quality monitoring, and future reporting design rather than direct operational scoring governance.

12.2 Current sub-score framework

Tester Quality sub-score	Current role	Main features used
Feedback Depth	Measures how detailed, specific, and actionable the tester's feedback appears.	Open-text word count, number of non-empty answers, specific issue terms, actionable feedback, vague-answer count.
Task Coverage	Measures whether enough task/component evidence exists to interpret the tester's contribution.	Attempted/evidenced tasks, task survey responses, transcript-evidenced tasks, insight task coverage, expected tasks.
Evidence Consistency	Measures whether survey, model, insight, and light manual-review signals broadly cohere.	Average raw survey score, Proposal 2 L3 average, friction counts, survey friction mentions, manual-review gap where available.

Tester Quality sub-score	Current role	Main features used
Engagement / Effort	Measures whether the tester appears to have contributed meaningful evidence without treating slowness as poor quality.	Survey completion, open-answer completion, attempted tasks, session duration, extremely short plus sparse-feedback pattern.

The current prototype combines four sub-score dimensions: Feedback Depth, Task Coverage, Evidence Consistency, and Engagement / Effort. Where a dimension is missing, the score is reweighted across the available dimensions rather than forcing a false zero.

12.3 Fairness safeguards, outputs and interpretation

Safeguard	Current treatment	Client-facing interpretation
Independence from ACC/USA	Tester Quality does not alter Accessibility or Usability scores.	It is a separate review signal, not a hidden multiplier.
Disability / AT / ESL protection	The methodology avoids direct penalties for disability status, assistive technology use, ESL status, or accessibility blockers.	The score should not be read as a judgement on participant capability.
Slow completion alone	Slow task duration alone is not treated as low quality.	Longer duration is not automatically poor-quality participation.
Manual review use	Manual review is used as a light consistency audit rather than definitive truth.	Tester Quality remains heuristic and suitable for calibration review.

The current methodology explicitly avoids direct penalties for disability, assistive technology use, slower reading, ESL status, accessibility blockers, or negative feedback. Slow task duration alone is not treated as low quality, and the manual-review comparison used inside the prototype is a light consistency audit rather than absolute ground truth.

A representative anonymised sample is shown below. Tester Quality is an independent third score and does not modify the Accessibility or Usability outputs. The full Tester Quality package accompanies the handover artefacts.

Participant	Quality score	Band	Feedback depth	Task coverage	Evidence consistency	Engagement / effort	Review flag
Participant A	98.38	high_quality	100.0	93.5	100.0	100.0	no_review_needed
Participant B	98.25	high_quality	100.0	100.0	93.0	100.0	no_review_needed
Participant C	95.75	high_quality	100.0	87.0	96.0	100.0	no_review_needed

Across the current package, the Tester Quality summary reports 21 testers, an average Tester Quality score of 90.12, and 18 testers in the current high-quality band.

12.4 Current limitations and future use

- The current Tester Quality outputs are separate from ACC/USA and should remain separate unless SMP chooses a stronger future role for them.
- The prototype is suitable for internal review and future development planning, but it is not yet positioned as a production reliability model.
- Any future operational use would benefit from a larger reviewed sample, clearer reporting requirements, and a decision on how the score should be surfaced to SMP stakeholders.
- The current artefacts include tester_quality_scores.csv, tester_quality_features.csv, tester_quality_summary.csv, tester_quality_methodology.md, tester_quality_case_examples.md, and the associated quality-check output.
- These artefacts are included as part of the broader analytical handover package rather than as replacements for the primary Accessibility and Usability outputs.

13. Manual Review Validation

13.1 Validation objective

The validation objective is to determine whether the current scoring outputs are more credible, more explainable, and better aligned with observed evidence than raw survey scores alone. The current validation approach is case-based and evidence-led rather than a claim of formal statistical ground truth.

13.2 Current validation evidence

Evidence used	Purpose in validation
Raw survey score	Shows the participant's original stated experience.
Model C score before time moderation	Shows the effect of current Proposal 1 / Model C scoring before Proposal 2.
Proposal 2 time-moderated L3 score	Shows the effect of time moderation where applicable.
Transcript-aligned task evidence	Supports case-based interpretation of whether the score appears justified, too harsh, too lenient, or too uncertain to judge confidently.
Insight evidence	Explains the friction type, severity, confidence, and task context used in scoring.
Reviewer judgement	Provides the current validation anchor while still preserving uncertainty where evidence is limited.

The current validation evidence package includes an 18-case manual-review sample, scoring rubric, error summary, findings notes, case examples, and quality-check outputs. It compares raw survey scores, Model C before time moderation, and Proposal 2 after time moderation where those values are available.

13.3 Interpretation approach

Model result	Behaviour evidence	Interpretation
Low model score	Strong aligned friction evidence	More likely to be a justified correction.

Model result	Behaviour evidence	Interpretation
Low model score	Weak aligned friction evidence	Potentially too harsh; review recommended.
High model score	Strong aligned friction evidence	Potentially too lenient; review recommended.
Proposal 2 time cap applied	Aligned time/transcript evidence supports friction	Time moderation appears more credible.
Proposal 2 time cap applied	Evidence remains mixed or noisy	Time signal should be treated cautiously and suitable for calibration review.
Insufficient aligned evidence	Task or participant matching remains incomplete	Retain as uncertain rather than forcing a conclusion.

The current results are best interpreted as early calibration evidence. They are useful for understanding where the scoring logic appears credible, too harsh, too lenient, or uncertain, but they do not yet support strong general claims about final production performance.

13.4 Current validation package

Case type	Why include it	Suggested interpretation label
Changed-score case	Checks whether a visible score reduction appears justified by aligned evidence.	Justified / too harsh / too lenient
Cap-triggered but unchanged case	Checks whether the cap was meaningful even when the score did not move.	Cap confirmed / cap not informative
Control case	Checks whether normal or positive experiences remain relatively stable.	Reasonable stability / over-correction
Missing-time-evidence case	Checks how incomplete time alignment should be handled.	Insufficient evidence / mapping issue
Verbose but successful case	Checks whether long duration can be noisy rather than genuinely friction-driven.	Time signal noisy / review recommended

The current package contains changed-score cases, cap-triggered-but-unchanged cases, control cases, and missing-time-evidence cases. Insufficient-evidence cases are preserved rather than forced into a definitive label, which is consistent with the overall conservative design of the handover package.

The main validation artefacts are outputs/manual_review_outputs/manual_review_cases.csv, outputs/manual_review_outputs/manual_review_scoring_rubric.csv, outputs/manual_review_outputs/manual_review_validation_summary.csv, outputs/manual_review_outputs/manual_review_error_details.csv, outputs/manual_review_outputs/manual_review_findings.md,

outputs/manual_review_outputs/manual_review_case_examples.md, and outputs/manual_review_outputs/manual_review_quality_check.md.

A concise validation summary is shown below. This is early calibration evidence rather than definitive ground truth, and the full manual review validation package accompanies the handover artefacts.

Compared output	Mean absolute error	Median absolute error	Evaluated cases	Closest score (%)
Raw survey score	13.57	10.0	14	78.6
Model C before time	13.28	10.0	14	85.7
Proposal 2 after time	19.49	10.0	14	71.4

Representative reviewed case: AAMI Website for Participant A. The raw survey score was 100.0, the pre-time usability score was 90.1, the Proposal 2 after-time score was 50.0, and the manual review label was moderate_friction. The reviewer summary described the case as: Task evidence shows real navigation and readability friction, including difficulty navigating headings and interpreting dense homepage content, but not a clear total tas…

14. Output Structure and Auditability

14.1 Implemented Output and Audit Principles

- The output structure includes intermediate scores, not only final scores.
- Score reductions are supported by explanation fields written in plain language.
- Time moderation effects are traceable from pre-time score to applied cap and final score.
- Unmatched tasks or participants are preserved in diagnostic files rather than being silently dropped.
- The delivered outputs support both technical review and stakeholder interpretation.

14.2 Core output fields

Field / field group	Purpose
participant / email / participant_key	Join participant across survey, insight, Proposal 2, validation, and Tester Quality evidence.
task_id / task_title / model_c_task	Join task-level scoring rows to time, validation, and reporting artefacts.
survey_question / raw_score / ceiling_score	Preserve the survey-derived starting reference.
primary_cohort / acc_weight / usa_weight	Record cohort-aware weighting assumptions used in the implementation.
l1_score	Show the categorical/accessibility stage before L2/L3.
l2_score / acc_score_final	Show Accessibility score before and after final safeguard logic.

Field / field group	Purpose
l3_score_before_time / l3_score_after_proposal2_time	Show Usability before and after Proposal 2 moderation where applicable.
acc_explanation / usa_explanation	Provide plain-language reasons for score stability or score reduction.
proposal2_time_cap / proposal2_changed_score / time_signal / confidence / proposal2_time_explanation	Record the time moderation decision, supporting signal, and explanation.

The current outputs are organised in groups: Proposal 1 scoring outputs, Proposal 2 moderation outputs, manual review validation outputs, Tester Quality outputs, rollup outputs, and data dictionary/methodology files.

14.3 Files register

Output group	File / artefact	Purpose and handover use
A. Proposal 1 outputs	outputs/proposal1_rows.csv	Primary row-level Proposal 1 scoring evidence.
A. Proposal 1 outputs	outputs/proposal1_df_insights.csv	Processed insight evidence table used in scoring.
A. Proposal 1 outputs	outputs/model_c3_proposal1_results.xlsx	Workbook-format Proposal 1 output for review.
A. Proposal 1 outputs	outputs/proposal1_task_summary.csv; outputs/proposal1_cohort_summary.csv; outputs/proposal1_project_summary.txt	Aggregate Proposal 1 summaries for reporting.
B. Proposal 2 outputs	outputs/l3_score_before_time.csv	Reference usability scores before Proposal 2 time moderation.
B. Proposal 2 outputs	outputs/proposal2_integrated_l3_time_score.csv	Primary integrated Proposal 2 row-level output.
B. Proposal 2 outputs	outputs/proposal2_time_audit_log.csv	Row-level audit trace for time moderation.
B. Proposal 2 outputs	outputs/proposal2_integration_summary.csv; outputs/proposal2_human_readable_summary.txt	Summary-level Proposal 2 review artefacts.
B. Proposal 2 outputs	outputs/proposal2_time_signal_result_recomputed_for_integration.csv; outputs/proposal2_unmatched_participants.csv; outputs/proposal2_unmatched_tasks.csv	Supporting diagnostics and recomputed time evidence.

Output group	File / artefact	Purpose and handover use
C. Manual review validation outputs	outputs/manual_review_outputs/*	Current validation sample, rubric, findings, error summary, and quality checks.
D. Tester Quality outputs	outputs/tester_quality_outputs/*	Independent Tester Quality prototype scores, features, methodology, and quality checks.
E. Rollup outputs	outputs/proposal2_rollup_task.csv; outputs/proposal2_rollup_participant.csv; outputs/proposal2_rollup_project.csv	Task-, participant-, and project-level summary outputs.
F. Data dictionary / methodology files	final_data_dictionary_package/*; README_How_To_Run.md	Field-level reference and practical run guidance for future maintenance.

The full files accompany the repository package, while the embedded examples in this document remain concise, anonymised, and selected for explanatory value.

Representative anonymised samples are included in this document, while the complete output files accompany the handover package. This allows reviewers to inspect examples in the document and trace the full artefacts in the repository package.

15. Retry Behaviour

Retry remains an exploratory behavioural signal in the current handover package. It may help explain prolonged task duration or repeated effort during review, but it is not currently used as a direct scoring rule in the delivered pipeline.

16. Limitations and Risks

Risk / constraint	Impact	Mitigation / documentation response
Notebook-driven workflow is not production integration	Current outputs are reviewable and reproducible, but not yet operationalised as a production SMP service.	Treat the package as a working analytical prototype and handover guide.
Time evidence coverage may be incomplete	Some participant/task rows will not have matched Proposal 2 evidence.	Preserve unmatched diagnostics and avoid implying full time coverage.
Time is noisy	Long duration may reflect participant style, assistive technology use, or task complexity rather than hidden friction alone.	Use Proposal 2 cautiously and continue calibration review.

Risk / constraint	Impact	Mitigation / documentation response
Manual review sample is limited	Validation insights are useful but not statistically definitive.	Treat validation as early calibration evidence.
Tester Quality is heuristic	The third score should not be overinterpreted as a production reliability model.	Keep it separate from ACC/USA and review future reporting needs.
LLM-derived insights are supporting evidence	Insight rows may miss non-verbal or non-verbalised barriers.	Keep confidence thresholding and auditability in place.
Thresholds and weights require governance	Future adjustments could materially change score behaviour.	Record any approved changes in both code and documentation.
Retry is not operationalised as a direct rule	Supporting behavioural interpretation may remain inconsistent.	Formalise retry only if SMP chooses a later implementation phase.
Privacy and anonymisation remain essential	Repository outputs may contain identifiable fields.	Use anonymised or client-approved examples in any outward-facing material.

17. Recommendations and Next Iteration Roadmap

17.1 Immediate handover recommendations

1. Convert the notebook sequence into a more repeatable operational pipeline once SMP decides the preferred future deployment pathway.
2. Expand the manual-review validation sample so threshold calibration and comparative claims can be reviewed against a broader evidence base.
3. Calibrate Proposal 2 time thresholds and confirm how raw task-duration sources should be governed in future reporting.

17.2 Future development priorities

Priority	Recommendation	Expected value
High	Productionise the notebook workflow into a more repeatable pipeline once the preferred operational path is agreed.	Improves repeatability and readiness for future integration.
High	Expand the manual-review validation sample.	Supports more confident calibration and interpretation.
High	Calibrate Proposal 2 time thresholds and benchmark assumptions.	Improves the reliability of time-based moderation.

Priority	Recommendation	Expected value
Medium	Connect and validate raw task-duration sources more directly.	Reduces matching and coverage uncertainty.
Medium	Decide how Tester Quality should be surfaced in SMP reporting.	Clarifies whether the third score remains internal or becomes part of future reporting.
Medium	Formalise retry extraction only if SMP wants retry operationalised.	Prevents premature use of ambiguous behavioural signals.
Medium	Add stronger automated tests across scoring and output generation.	Improves change control and reproducibility.
Longer term	Create client-approved output templates and governance review for threshold changes.	Supports clearer future handover and reporting consistency.

18. Handover Checklist

The handover package includes the client-facing documentation, notebooks 01–06, the outputs folder, the manual_review_outputs folder, the tester_quality_outputs folder, the final_data_dictionary_package, run guidance, representative anonymised examples in this document, and the handover artefact register.

Included artefact	Status	Notes
Client-facing documentation	Included in handover package	Primary client-facing explanation of the delivered scoring package.
Notebooks 01–06	Included in repository package	Provide the notebook-driven implementation, rollups, validation, and presentation assets.
Proposal 1 output artefacts	Included in output folder	Provide row-level Proposal 1 scoring outputs and summaries.
Proposal 2 output artefacts	Included in output folder	Provide integrated time moderation outputs, audit logs, diagnostics, and rollups.
Manual review validation outputs	Included in validation package	Provide the 18-case review sample, rubric, findings, and summary metrics.
Tester Quality outputs	Included in Tester Quality package	Provide the independent third-score prototype outputs and methodology notes.

Included artefact	Status	Notes
Data dictionary package	Included in repository package	Provides field definitions, schema documentation, and file inventory.
Run guidance	Included in repository package	Documents notebook order, prerequisites, and output dependencies.
Handover artefact register	Included in document appendices	Lists the accompanying artefacts and packaging notes.
Known limitations and future work summary	Included in document	Summarises production integration considerations and future development priorities.

Future operational decisions include whether SMP wants additional stand-alone reporting examples, whether a document-only circulation format would require extra embedded artefacts, any formal sign-off materials required for issue, and any preferred packaging changes for external circulation.

Appendix A. Glossary

Term	Meaning in this project
ACC	Accessibility score or accessibility scoring stream.
USA	Usability score or usability scoring stream.
AT	Accessibility / assistive technology friction stream.
COMP	Comprehension friction stream.
L1	The first scoring layer, primarily used for severe or categorical accessibility rules and hard caps.
L2	The second scoring layer, used for accessibility and comprehension adjustments before final accessibility scoring.
L3	The third scoring layer, used for usability adjustments and the Proposal 2 time moderation step.
Friction	Evidence of difficulty, confusion, delay, or blockage observed in the participant experience.
Cap	A maximum score limit applied when evidence indicates the score should not remain above a defined level.

Term	Meaning in this project
Floor	A minimum score boundary used when a deduction applies, so the score does not fall below the defined implemented lower limit for that rule.
Proposal 1	Graduated medium-friction deduction layer inside current Model C/C3 scoring.
Proposal 2	Participant-normalised time moderation layer for L3 usability.
Tester Quality Rating	Separate third-score prototype used for feedback/evidence-quality review.
Artefact	A delivered file, output, or supporting package included with the handover materials.
Repository package	The accompanying project folders, notebooks, outputs, and supporting files delivered with this handover document.
Anonymised sample	A shortened example in which participant identifiers and sensitive details are removed or masked for client-facing use.
Ceiling score	Survey-derived starting reference that the model can reduce but generally should not exceed.
Audit field	Output field that explains or traces why a score changed or stayed the same.

Appendix B. Scoring Rules and Thresholds

B.1 Proposal 1 graduated deduction constants

Rule group	Base penalty	Floor	Applies to
Medium AT	20	30	Medium accessibility friction.
Medium COMP	15	50	One or two medium comprehension friction events.
Medium USA single/combined	12	75	One or more medium usability friction events, subject to stream weighting in the current implementation.
Medium USA stacking	25	70	Three or more medium usability friction events.

This appendix summarises the main implemented rules and thresholds used in the current repository. The rule tables are intended as implementation references for the handover package, not as final client-approved governance settings. In these tables, a floor is the implemented lower score boundary for a rule once a deduction has been triggered.

B.2 Hard rules retained

Rule	Effect	Reason retained
Blocker or severe accessibility friction	Score = 0	Categorical failure / participant could not meaningfully engage.
One high accessibility friction	Cap 20	High access barrier must be strongly reflected.
Two or more high accessibility friction events	Cap 10	Repeated high access barriers indicate severe exclusion risk.
Severe usability or 3+ high usability friction	Cap 50	Severe usability friction remains materially important.
High + medium usability friction	Cap 60	Stacked friction remains too serious for a mild deduction only.
Medium comprehension x3+	Cap 40	Stacked comprehension friction logged for future review and not graduated in this phase.

Where a rule remains intentionally conservative, that has been noted in the rationale column. The same approach should be maintained in any future productionisation work unless threshold governance is updated formally.

B.3 Proposal 2 time cap bands

Normalised ratio band	Cap	Calibration note
2x-3x	50	Implemented in current code; retained as the current implemented setting and suitable for future calibration review if SMP wants recalibration.
3x-4x	40	Implemented in current code; retained as the current implemented setting and suitable for future calibration review if SMP wants recalibration.

Normalised ratio band	Cap	Calibration note
4x+	25	Implemented in current code; retained as the current implemented setting and suitable for future calibration review if SMP wants recalibration.

The Proposal 2 ratio bands and caps shown here are the current implemented settings. They are useful for transparent review and for future calibration planning.

Any later threshold adjustments should be reflected simultaneously in the notebooks, output explanations, and threshold documentation so the handover package remains internally consistent.

Appendix C. Repository Reference Register

The repository reference register below lists the main notebooks, outputs, and supporting packages used to prepare and interpret the current handover package.

Repository reference	Use in this handover pack
README_How_To_Run.md	Used for current execution order, repository structure, and stated workflow constraints.
01_run_model_c_proposal1.ipynb	Used for Model C / Proposal 1 logic, constants, and output references.
02_proposal2_time_integration.ipynb	Used for Proposal 2 timing logic, thresholds, diagnostics, and integration outputs.
03_manual_review_validation.ipynb	Used for current manual-review validation package design and outputs.
04_tester_quality_rating.ipynb and nb04_tq_script.py	Used for current Tester Quality prototype logic and outputs.
05_presentation_assets.ipynb	Used for presentation and visual asset outputs.
06_proposal2_rollup_outputs.ipynb	Used for task-, participant-, and project-level rollup outputs.
outputs/ folder	Used as the primary output evidence referenced throughout the document.
final_data_dictionary_package/	Used as the field-level and file-level reference package.

Interpretation note: the current repository package is the primary implementation reference for this handover pack. Background project materials remain useful context, but the scoring behaviour described here reflects the current notebooks and delivered outputs.

Appendix D. Accompanying Handover Artefact Register

Artefact	Purpose	Location	Status	Notes
Proposal 1 output package	Row-level Proposal 1 scoring evidence plus task, cohort, and project summaries.	outputs/proposal1_rows.csv and related Proposal 1 files	Included in output folder	Core scoring evidence for Proposal 1 review.
Proposal 2 moderation package	Integrated Proposal 2 outputs, time audit logs, diagnostics, summaries, and rollups.	outputs/proposal2_integrated_l3_time_score.csv and related Proposal 2 files	Included in output folder	Core moderation evidence for Proposal 2 review.
Manual review validation package	Validation sample, rubric, findings, summary metrics, and quality checks.	outputs/manual_review_outputs/	Included in validation package	Supports early calibration review.
Tester Quality package	Independent third-score prototype outputs, methodology, examples, and quality check.	outputs/tester_quality_outputs/	Included in Tester Quality package	Supports feedback-quality interpretation and future development.
Data dictionary package	Field definitions, output schema documentation, and file inventory references.	final_data_dictionary_package/	Included in repository package	Field-level reference for technical review.
Run guidance	Notebook order, prerequisites, and output dependency	README_How_To_Run.md	Included in repository package	Supports future maintenance and reruns.

Artefact	Purpose	Location	Status	Notes
	guidance.			
Presentation assets	Charts and visual assets supporting reporting and presentation.	outputs/presentation_assets/	Included in repository package	Optional supporting visuals.
Output schema documentation	Schema-level references for principal delivered output files.	final_data_dictionary_package/output_schema_documentation.csv	Included in repository package	Useful for technical interpretation of delivered outputs.